

```

import pandas as pd
import numpy as np
np.random.seed(42)
data = {
    "Age": np.random.randint(21, 60, 100),
    "Salary": np.random.randint(30000, 120000, 100),
    "Experience": np.random.randint(1, 35, 100),
    "Department": np.random.choice(["HR", "IT", "Sales", "Finance"],
100),
    "Gender": np.random.choice(["Male", "Female"], 100),
    "PerformanceScore": np.random.randint(1, 11, 100)
}
df = pd.DataFrame(data)
print(df.head())

```

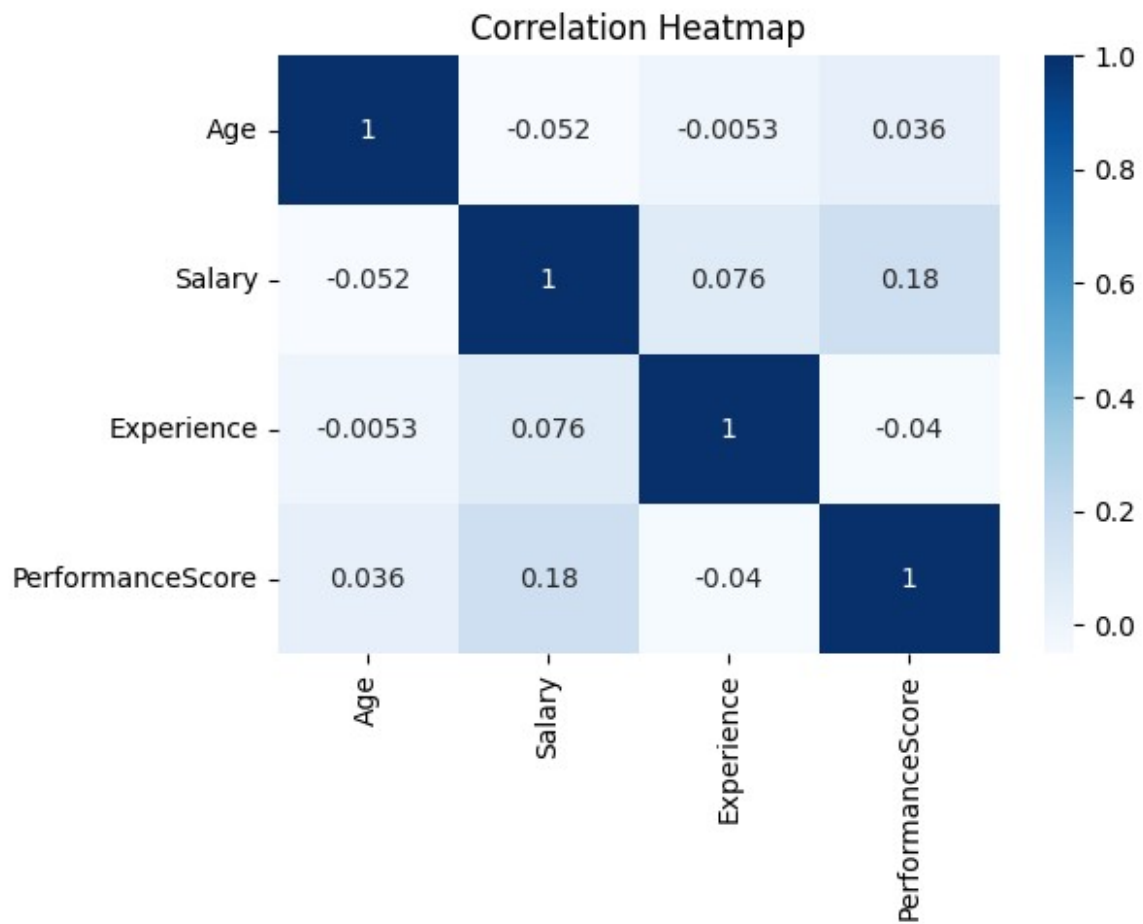
	Age	Salary	Experience	Department	Gender	PerformanceScore
0	59	114651	6	HR	Male	3
1	49	93335	16	Finance	Female	1
2	35	40965	29	HR	Male	1
3	28	54538	3	HR	Male	8
4	41	100592	20	Sales	Female	10

```

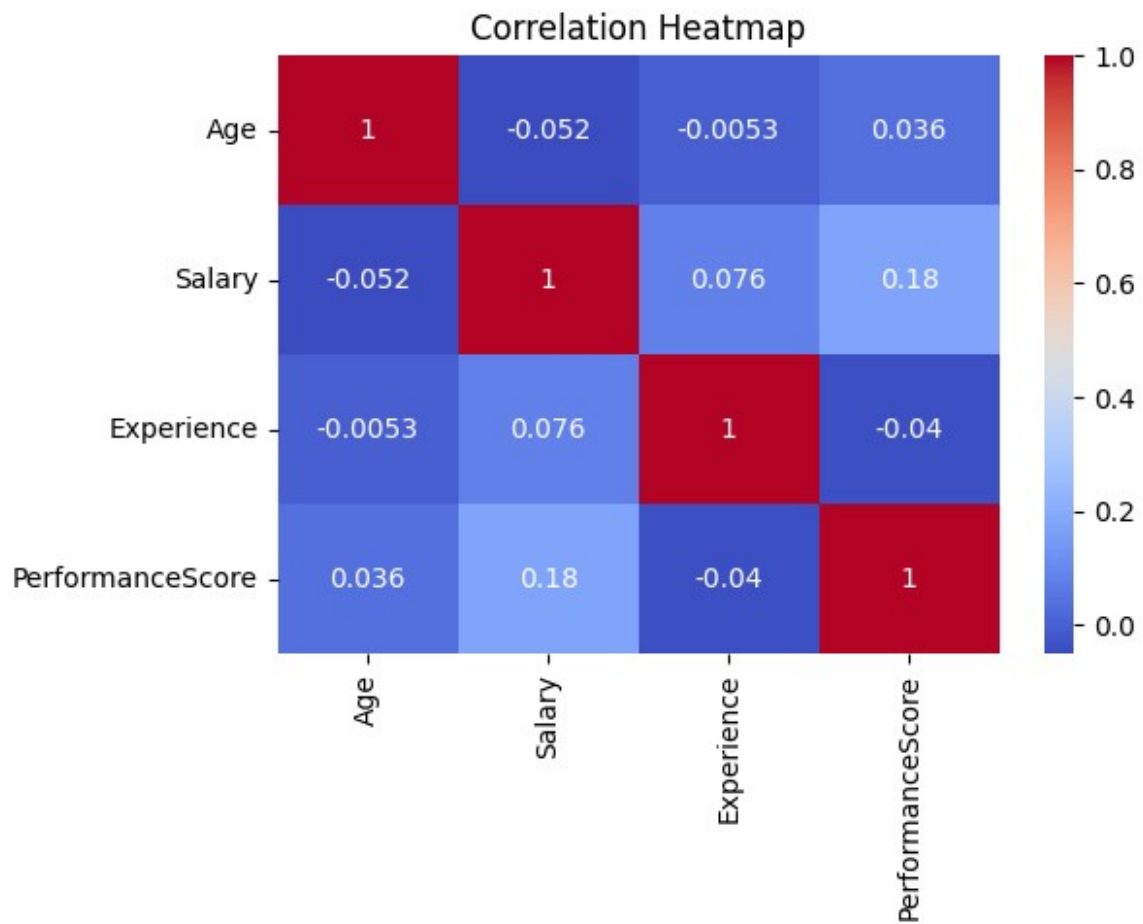
import seaborn as sns
import matplotlib.pyplot as plt

#correlation heat map
plt.figure(figsize=(6,4))
corr=df.corr(numeric_only=True)
sns.heatmap(corr,annot=True,cmap="Blues")
plt.title("Correlation Heatmap")
plt.show()

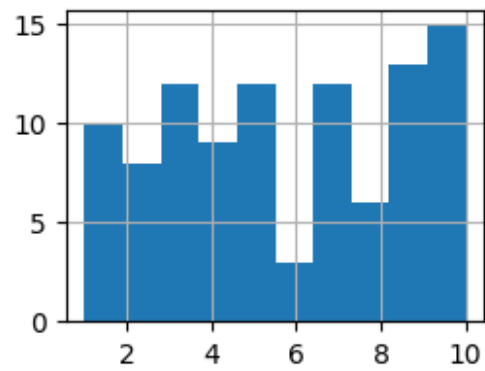
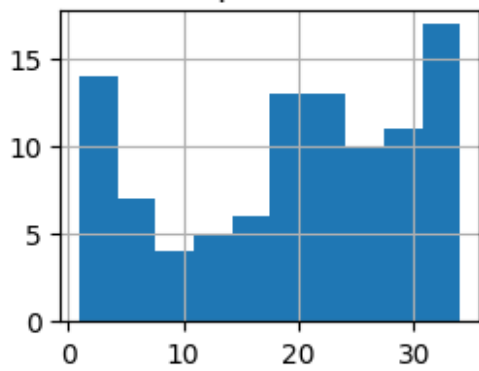
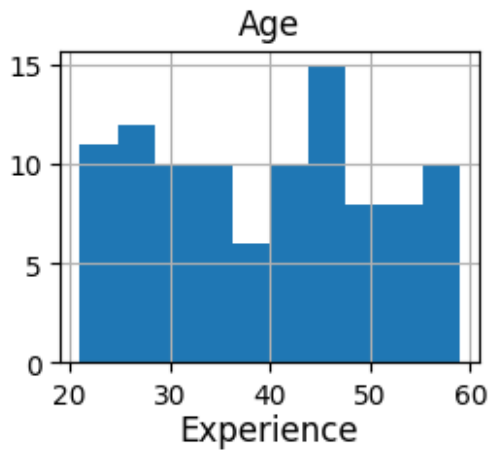
```



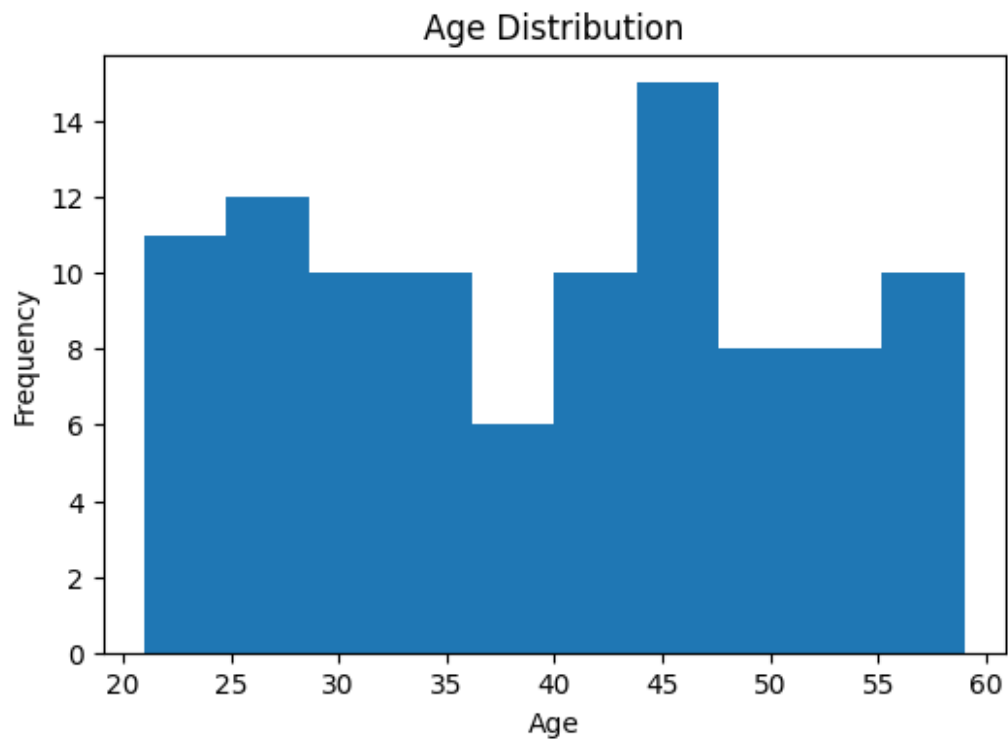
```
#correlation heat map
plt.figure(figsize=(6,4))
corr=df.corr(numeric_only=True)
sns.heatmap(corr,annot=True,cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



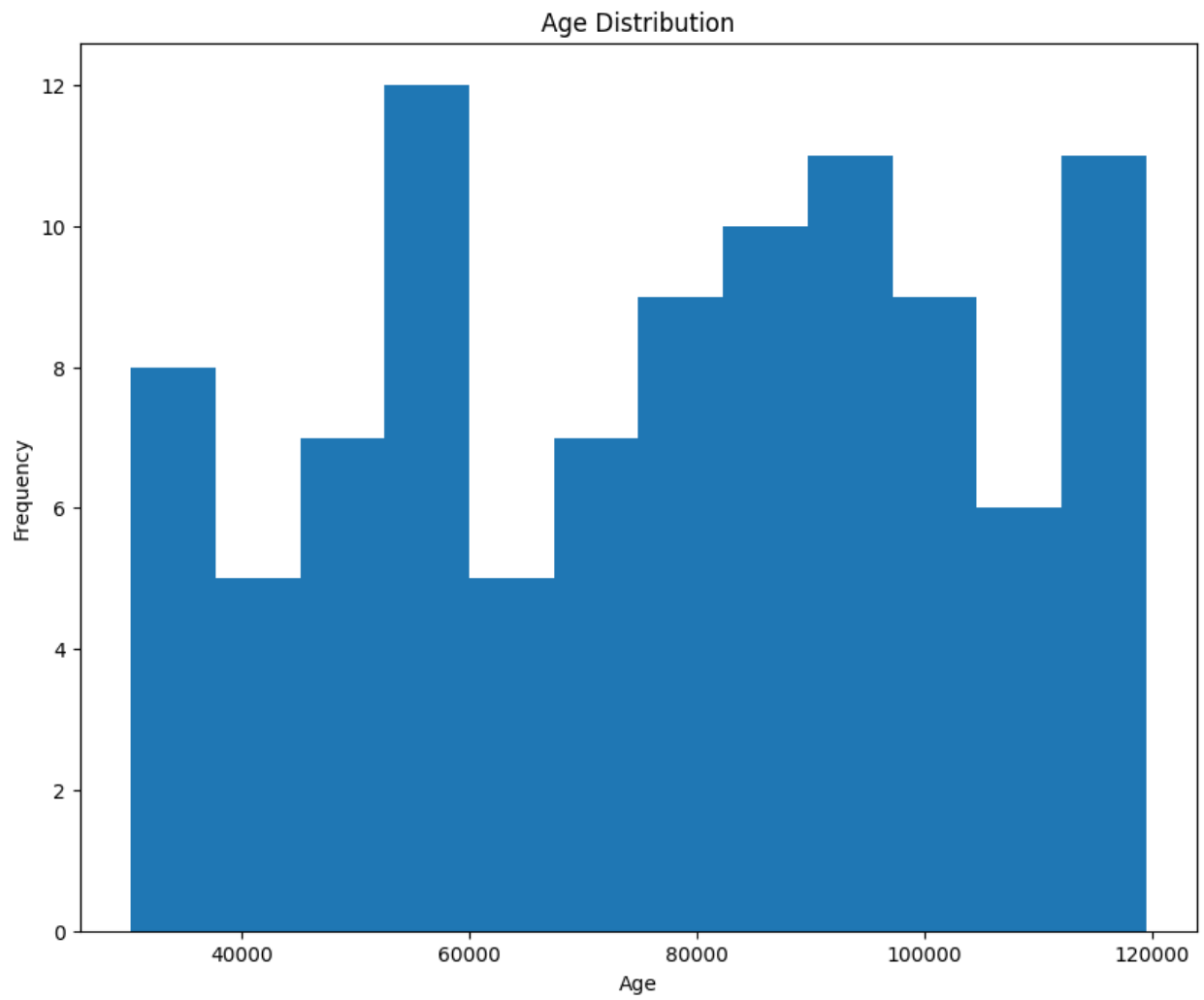
```
df.hist()  
array([[<Axes: title={'center': 'Age'}>,  
       <Axes: title={'center': 'Salary'}>],  
       [<Axes: title={'center': 'Experience'}>,  
       <Axes: title={'center': 'PerformanceScore'}>]], dtype=object)
```



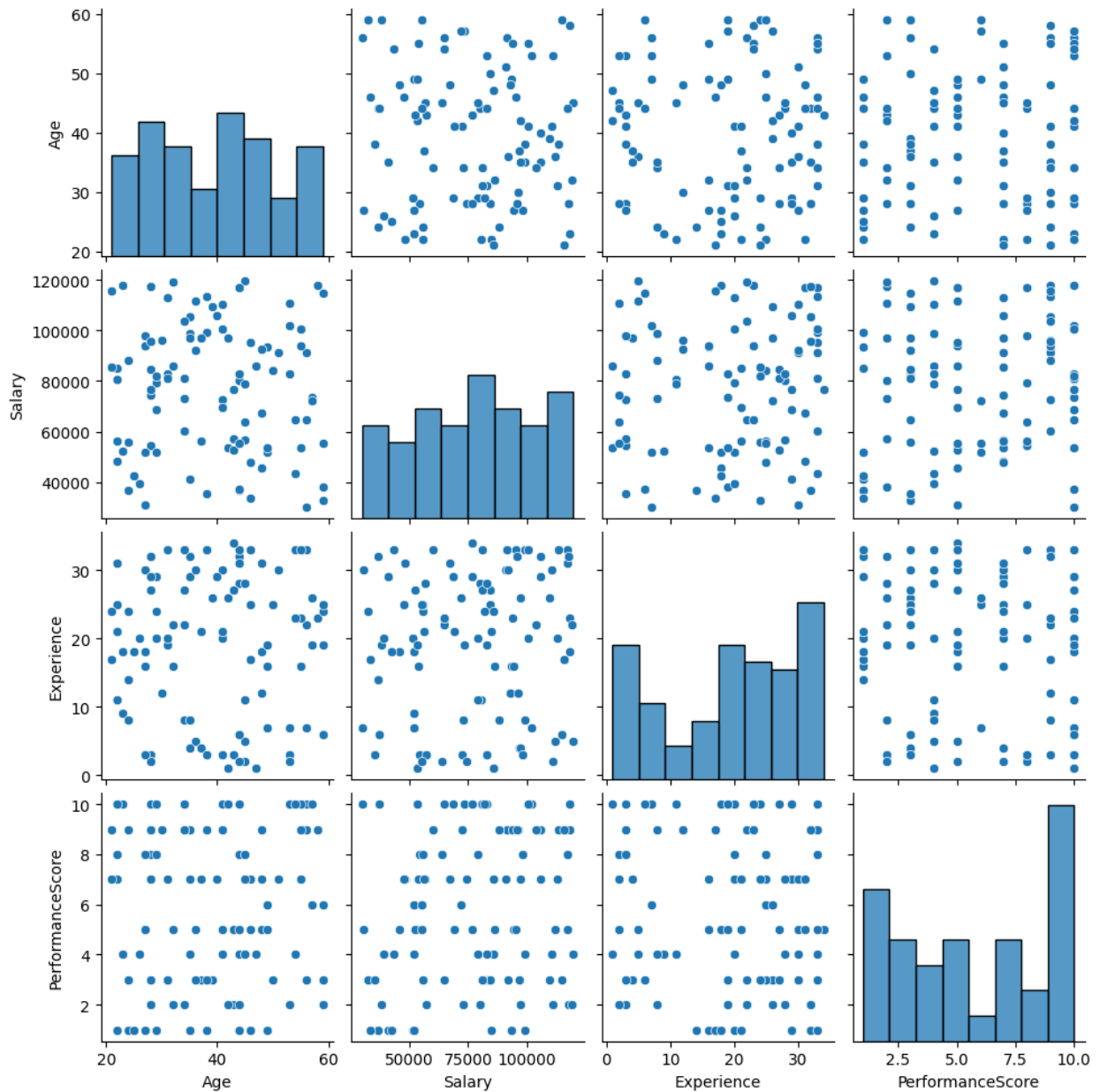
```
plt.figure(figsize=(6, 4))
plt.hist(df["Age"], bins=10)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



```
plt.figure(figsize=(10, 8))
plt.hist(df["Salary"], bins=12)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



```
sns.pairplot(df)  
plt.show()
```



```
plt.figure(figsize=(6,4))
sns.scatterplot(x="Experience",y="Salary",data=df)
plt.title("Experience vs Salary")
plt.show
```

```
<function matplotlib.pyplot.show(close=None, block=None)>
```



```
plt.figure(figsize=(6,4))  
sns.violinplot(x="Department",y="Salary",data=df)  
plt.title("Salary Distribution (Violin Plot)")  
plt.show()
```




```
plt.figure(figsize=(6,4))
sns.boxplot(x="Department",y="Salary",data=df)
plt.title("Salary Distribution ")
plt.show()
```

